

TECHNICAL DATA SHEET

AF55 Series**
HYDROPLUS™ WATERBORNE CRYSTAL CLEAR TOPCOAT

Custom Blends: A55XX Series

Waterborne Crosslinker (optional): XA4081000

5 Gloss Clear..... AF5505000

20 Gloss Clear..... AF5520000

50 Gloss Clear..... AF5550000

Gloss level:	05, 20, 50 gloss
Area of use:	Interior wood products such as residential furniture, cabinetry, architectural woodworking
Method of use:	Spray - Conventional, Airless, Air Assisted Airless, HVLP.
Mixing Procedure:	Ready to use. If thinning is required, use deionized water up to 5%.
Clean up:	2 parts water to 1 part Butyl Cellosolve

Technical Characteristics

Solids content (%):	Weight Solids 21.9 ± 1 / Volume Solids 18.7 ± 1
Specific gravity:	1.03
Viscosity (Zahn #3 at 77°F):	17-22 seconds
Recommended film thickness (per coat) :	4.0 – 5.0 mils wet / 0.75 – 1.0 mils dry
Recommended number of coats:	3-4 (Do not exceed 4.0 mils dry for complete system)
Spread Rate:	300 average sq. ft/gal @ 1.0 mils DFT, No loss
Drying time (at 77°F):	Dust free 20 minutes Touch dry 30 minutes Sand 30 – 45 minutes Recoat 30 – 45 minutes Stackable 24 hours
Forced drying (80 g/m²):	Flash-off 15 minutes Hot air @ 95°F 45 minutes Cooling 15 minutes
Shelf-life:	12 months After long periods of storage, always check homogeneity and stir well before use to eliminate any possible sediment.
VOC (Theoretical as packaged, maximum):	1.85 lb/gal, 221 g/L (less exempt solvents)
VHAPS (as packaged, maximum):	0.09 lb/lb of solids
Flash Point	N.A.
An Environmental Data Sheet is available from your local servicing facility, or at www.paintdocs.com .	

General Characteristics

HYDROPLUS™ Waterborne Crystal Clear Topcoat is a fast drying waterborne, acrylic finish for interior wood products that ensures good film hardness, transparency and smoothness to the touch. This clear finish can be used as a self-sealing system or over HYDROPLUS™ Waterborne Basecoat, AU0469. This finish can be catalyzed to improve film hardness and chemical resistance performance. Typical end use applications include interior wood products in the residential furniture, architectural millwork and novelty segments.

Substrate Preparation

Wood (interior only): Must be clean, dry and finish sanded. Substrate should be free of grease, oil, dirt, fingerprints and any contamination to ensure optimum adhesion and coating performance properties. Moisture of the wood should be 6-8%.

TECHNICAL DATA SHEET

AF55** Series

HYDROPLUS™ WATERBORNE CRYSTAL CLEAR TOPCOAT

Catalyzing

For increased film hardness and chemical resistance, HYDROPLUS™ Waterborne Crystal Clear Topcoat may be catalyzed at 1% by volume with a 50/50 blend of Waterborne (aziridine) Crosslinker and deionized water. Stir the blend well. Additional catalyst volumes will not improve performance properties. When catalyzed, this material has a 24 hour pot-life. You may carry over the catalyzed product overnight by adding 50% of the uncatalyzed coating at the end of the day. Catalyze the entire amount of material, not just the newly added material when ready to use the following day. Catalyzed material must not be carried over the weekend and should be properly disposed of after the second 24 hour period.

Wood Finishing System (self-sealer)

1. **Stain (optional)** – Stain or color wood if desired and allow to dry. Consult your local representative for appropriate stain offering.
2. **Seal** – Apply first coat of Hydroplus Waterborne Crystal Clear at 4.0-5.0 mils wet. Allow to dry for 30-45 minutes. Sand 320 grit sand paper. Remove sanding dust or debris. For best sanding results, use steared sand paper. Can use AU0469 Waterborne Basecoat as a sealer coat.
3. **Topcoat** – Apply Hydroplus Waterborne Crystal Clear at 4.0-4.5 mils wet. Allow to dry for 30-45 minutes. Sand with 320 grit sand paper. Remove sanding dust or debris.
4. **Topcoat** – Apply additional coat of Hydroplus Waterborne Crystal Clear at 4.0-5.0 mils wet.
5. For additional build or depth, apply additional topcoat of AF55 if desired. Do not exceed 4.0 mils total dry film thickness for complete finishing system.

Drying Conditions: This product dries primarily by water evaporation. Best drying occurs at 50% RH or lower and temperatures of 77°F or higher. Good air movement is essential for complete drying.

Special instructions:

- Keep from freezing. Do not store at temperature below 41°F.
- Do not apply at temperature below 59°F.
- Stir the product well before application in order to disperse any possible sedimentation. This operation is essential to ensure even matting on the substrate.
- Because of the variety of materials used to manufacture wood furniture, when moving to a waterborne system from a solvent based system, it is very important to check with the supplier the suitability of equipment and components used, with particular attention to electrostatic guns, pumps, seals, silicones, glues, products for booth water treatment, packing materials, sanding papers, putties, etc.
- The use of coating devices not in perfect order (defective gaskets, too high pressures) or of pumps with low capacity may cause major defects in the coating film (air bubbles).
- Waterborne Crosslinker (XA4081000) is a aziridine product. Thoroughly review product label and Safety Data Sheet (SDS) for safety information and cautions prior to using these products.
- Once the can has been opened, the coating may rot because of the attack of bacteria commonly present in the air. This phenomenon is easily detectable because of the bad smell and by an increase of viscosity of the product stored in the can. Avoid leaving containers open for long periods of time and reclaiming used coating into fresh product.

Testing: The information, data, and recommendations set forth in this Product Data Sheet are based upon test results believed to be reliable. However, due to the wide variety of substrates, substrate properties, surface preparation methods, equipment and tools, application methods, and environments, the customer should test the complete system for adhesion, compatibility and performance prior to full scale application.

TECHNICAL DATA SHEET

AF55 Series**

HYDROPLUS™ WATERBORNE CRYSTAL CLEAR TOPCOAT

Application Equipment Settings			
<u>Conventional</u>		<u>Air Assisted Airless</u>	
Air Pressure:	40 – 50 psi	Air Pressure:	10 - 15 psi
Fluid Pressure:	4 - 10 psi	Fluid Pressure:	400 – 600 psi
Cap/Tip Size:	.046 - .055	Cap/Tip Size:	.011 - .013
<u>Airless</u>		<u>HVLP</u>	
Pressure:	1400 - 1800 psi	Air Pressure @ Cap:	5 - 8 psi
Tip Size:	.011 - .015	Fluid Pressure:	4 - 8 psi
		Cap/Tip Size:	.046 - .055

FOR INDUSTRIAL SHOP APPLICATION ONLY

Thoroughly review product label and Safety Data Sheet (SDS) for safety information and cautions prior to using this product.

To obtain the most current version of the Environmental Data Sheet (EDS), Product Data Sheet (PDS), or Safety Data Sheet (SDS) please visit your local facility or www.paintdocs.com.

Please direct any questions or comments to your local servicing facility.

Note: All purchases of products from Sherwin-Williams are exclusively subject to Sherwin-Williams' terms and conditions of sale which can be found at www.sherwin.com. Please review these terms and conditions prior to the purchase of the products. Sherwin-Williams warrants the product to be free of manufacturing defect in accordance with Sherwin-Williams' quality control procedures. Except for the preceding sentence, due to factors that are outside of Sherwin-Williams' control, including substrate selection, and customer handling, preparation, and application, Sherwin-Williams cannot make any other warranties related to the product or the performance of the product. **SHERWIN-WILLIAMS DISCLAIMS ALL WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE IMPLIED WARRANTY OF MERCHANTABILITY, THE IMPLIED WARRANTY OF FITNESS FOR A PARTICULAR PURPOSE.** Liability for products proven to be defectively manufactured will be limited solely to replacement of the defective product or the refund of the purchase price paid for the defective product, as determined by Sherwin-Williams. Under no circumstances shall Sherwin-Williams be liable for indirect, special, incidental or consequential damages, lost profits or punitive damages arising from any cause whatsoever.